



RAPHAEL PELERIN

Software engineering · Epitech, class of 2027

I build across the stack, and most of what I have made lives where the layers meet — a front end bound by a scheduling constraint, an image pipeline that has to fit in 8 GB of VRAM, an audio plugin whose interface is drawn by the same code that reads the envelope. Five projects, four of them used by people who are not me.

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LANGUAGES

FR · EN bilingual · 中文 HSK3

Fifth-year software engineering student at Epitech. I would rather ship something narrow that works than something broad that almost does.

A
 AVAILABILITY

OPEN TO OFFERS

End-of-studies internship, 4–6 months
 Starting March 2027

B
 LANGUAGES

French
 English
 Mandarin
 Italian

NATIVE
BILINGUAL
HSK 3
BASIC

C
 EDUCATION

2022 – 2027	Epitech Technology Master's, Computer Engineering - Nice Functional and object-oriented programming, Unix, assembly, systems and network programming. Specialisations in DevOps, AI, web and mobile, and cybersecurity.
2025 – 2026	Beijing Jiaotong University Fourth year abroad - Beijing, China Software engineering, alongside Mandarin to HSK 3.
2022	Lycée Simone Veil General Baccalaureate Computer Science (NSI) and Mathematics.

D
 EARLIER

2023	ARD Mobile development intern An Angular application to manage NFC cards and user profiles for alarm systems, including card scanning to display cardholder information.
2022	Intel · AI for Youth Competition – 2nd place, team Computer vision and object recognition in Python with TensorFlow.

E
 CAPABILITIES

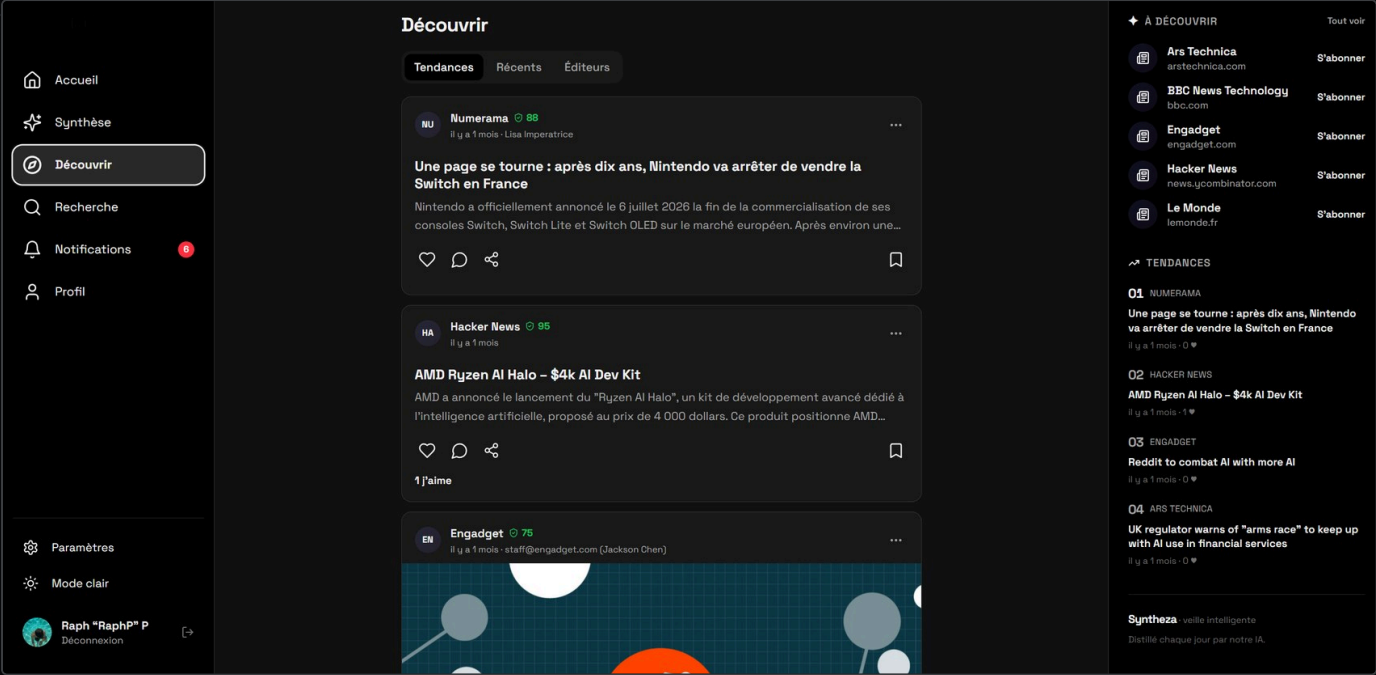
01	Web & mobile React · React Native · Angular · Ionic · Node.js · Express · Android · iOS
02	Backend FastAPI · Flask · Express · REST APIs · SOAP / XML
03	AI & generative OpenAI · Anthropic Claude · Gemini · LLM integration · TensorFlow · diffusion models
04	Data & infra PostgreSQL · MongoDB · MySQL · Prisma · Docker · containerisation · GitLab CI
05	Systems & audio C++ · JUCE · real-time DSP · Unix · assembly · systems and network programming

01 — LEAD FRONTEND WEB · EPITECH G-EIP-700 · 2025 — 2026

SYNTHEZA

Strategic watch, against information overload

LIVE [syntheza.ovh](#)

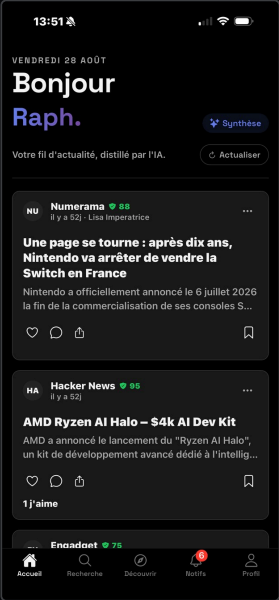


WEB CLIENT · DISCOVER FEED, PUBLISHERS, TRENDING AND TRUST FACTOR SCORES

A one-year entrepreneurial project run by three Epitech students working from three different countries. Syntheza ingests RSS at scale, deduplicates what it finds, and returns a daily synthesis instead of another feed. I lead the **web client** — the interface, the design system and the React component layer.

The green shield on every source is the **Trust Factor**: a corroboration score, so a claim made once does not read like one confirmed by five.

The team ran a field study of **20 professionals before writing code** — 85% reported daily overload, 70% lose one to two hours a day sorting noise. That study, not a hunch, set the scope.



MOBILE · EXPO

3

DEVELOPERS,
THREE COUNTRIES

~330 kB

GZIPPED
WEB BUNDLE

7

LANGUAGES
ON MOBILE

Live

IN PRODUCTION

React 19

TypeScript

Tailwind v4

Radix UI

Framer Motion

Vite

Node 20

Express

Prisma 7

PostgreSQL 15

Gemini 2.5 Flash

React Native · Expo

02 — SOLO PROJECT

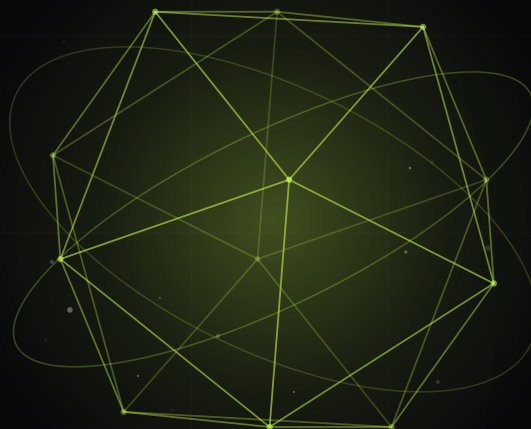
FORGE

Game-asset generation that never leaves the machine

Generating *one* good image with an AI tool is easy. Generating **forty that look like they came from the same game** is the actual problem — and it is a software engineering problem, not a prompting one.

Forge solves it with **recipes**: presets that own the style, resolution, sampler and post-processing for a category of asset. The user supplies only the subject, so two characters generated six months apart go through an identical pipeline and match.

It runs entirely on a local GPU. No API, no subscription, no account, and **no image ever leaves the machine**. Nine-slice UI panels are not trusted to come out symmetrical — the tool keeps one corner, mirrors it, and *méasures* the border thickness by scanning inward until the colour goes flat.



FOUR THINGS IT TOOK GENERATING IMAGES TO FIND

- 01 The text encoder only reads English. French subjects failed **6 out of 6**; English succeeded 6 out of 6. Fixed with a local translation model.
- 02 The **75-token prompt budget overflows silently** — the end is dropped with no error. A test now stops any recipe exceeding it.
- 03 Style strength has to vary **by asset type**: pushed hard, icons collapse into unreadable clusters at their real 48 px display size.
- 04 On 8 GB of VRAM an extra refiner pass took generation from **21 seconds to over 9 minutes**. Disabled by default, with the measured figure.

~4,100

LINES OF
PYTHON + JS

59

TESTS, ~5 S,
NO GPU NEEDED

~22 s

PER GENERATED
IMAGE

0

EXTERNAL API CALLS
AT RUNTIME

PyTorch 2.13

CUDA 12.6

diffusers · SDXL

ControlNet

LoRA

FastAPI

Pydantic v2

pytest

React 18

Ant Design 5

Vite 6

03 — SOLO PROJECT · 2025

KEYS

A VST3 sampler, drawn by hand



KEYS · SAMPLE LOADED, LIVE ADSR CURVE AND FILTER RESPONSE

A single-sample VST3 instrument built on JUCE. It pitch-shifts one sample across the keyboard through an ADSR envelope, a three-mode filter whose cutoff can follow that envelope, and forward or ping-pong looping across **1 to 16 voices**.

The interface uses **no GUI library at all** — every knob, curve and meter above is drawn with the JUCE Graphics API. The envelope view shows the ADSR shape with a live line tracking the playing voice; the filter graph draws the real frequency response with separate markers for the static and the envelope-modulated cutoff. The waveform is held as a **min/max envelope rather than raw samples**, so it stays sharp at any zoom, and the trim and loop markers are dragged directly on it.

16

SIMULTANEOUS
VOICES

3

FILTER MODES,
ENVELOPE-MODULATED

0

EXTERNAL GUI
LIBRARIES

VST3

WINDOWS X64,
FULLY AUTOMATABLE

C++

JUCE 8

Real-time DSP

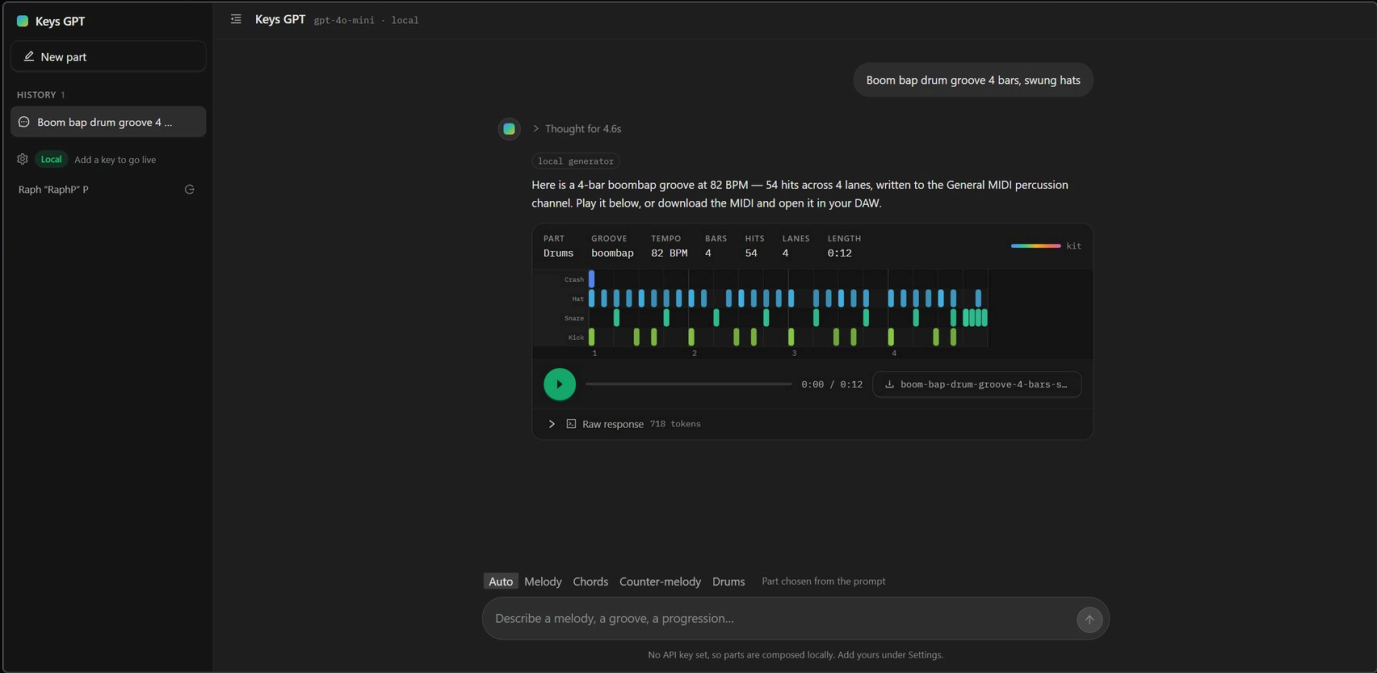
IIR filters

Projuicer

Visual Studio 2022

KEYS GPT

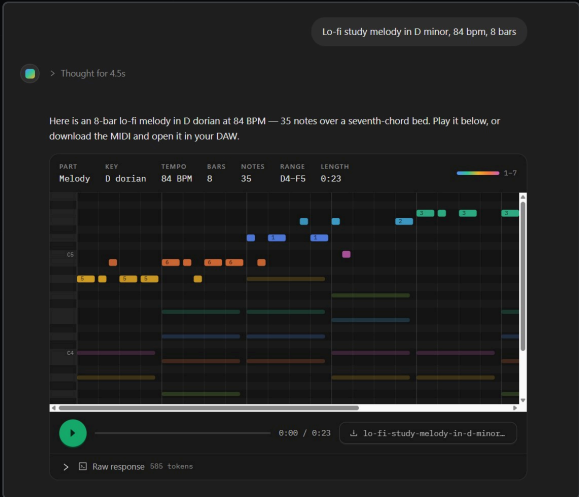
Describe a part, get a MIDI file



GENERATED 4-BAR BOOMBAP GROOVE · 54 HITS, FOUR LANES, PLAYABLE IN THE BROWSER

You describe a part — a mood, a key, a tempo, a length — and get a piano roll you can hear and a MIDI file you can keep. It writes melodies, chord progressions, counter-melodies and drum grooves, and the part type is inferred from the prompt rather than picked from a menu.

It runs against the **OpenAI API** — gpt-4o-mini above — and the part is playable in the browser before anything is downloaded, so you hear it before deciding whether to keep it.



8-BAR MELODY · D DORIAN, 35 NOTES

5

PART TYPES,
INFERRED FROM PROMPT

MIDI

DOWNLOADABLE,
OPENS IN ANY DAW

OpenAI

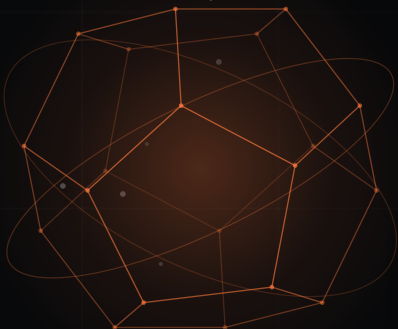
API
COMPATIBLE

APPOLONIA

Casino floor management, where a mistake takes a machine offline

Appolonia builds the systems casinos use to run their gaming floors. I worked on **KAMA**, the manager operators use to configure and deploy to **Kalypse Touch** terminals — the screens embedded in slot machines.

My responsibilities went from a dashboard widget, to redesigning whole modules, to building features from nothing.



REDESIGN

KT Updater

The module worked but was hard to use, and deployments failed often. I rebuilt it as three explicit steps — upload the file, choose whether it triggers a full reboot or an interface restart, then pick the devices and the schedule. Drag-and-drop uploads, live progress, and options that only appear when they apply.

In real deployments; used regularly by casino staff.

BUILT FROM SCRATCH

Administrator panel

Power policies for groups of terminals — "shut down every Friday at midnight", "lock from 2 AM to 6 AM daily" — with manual override for when the floor disagrees with the schedule.

BUILT FROM SCRATCH

Email update tool

Client update mails were written from scratch every time. A structured form now generates them in one consistent format, with a preview before sending.

ALSO

The "Today's Campaigns" dashboard widget, plus ongoing bug fixing, responsive layout work and component refactoring — with notes left in the code explaining why.

Angular 13

Ionic 6

TypeScript

SOAP / XML APIs

GitLab

F — AWAY FROM THE SCREEN

Skiing and taekwondo. Motorsport. Music production in a DAW — which is where the sampler came from. Travel, most recently a year of it in Beijing.

Full driving licence.

G — AVAILABILITY

● OPEN TO OFFERS

End-of-studies internship, 4–6 months
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GET IN TOUCH

LET'S BUILD SOMETHING SOLID

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BASED IN

Biot, France · 06